Validation and AI-Use Record

Tesla, Inc. (TSLA) | Frozen Project 1 record | October 2026

AI-use disclosure

Codex was used for project structuring, code assistance, research leads, and explanatory support. Its output was treated as a lead or draft only. A preliminary 5.26% risk-free rate was checked against the official Federal Reserve H.15 series, corrected to 5.29%, and rerun through the model. The disposition is qualified and corrected; AI-generated judgment was not accepted without cited evidence.

Validation register

Check	Pass criterion / result	Evidence
Known-answer reconciliation	EV $39,158.52m; equity $74,841.52m; $21.21/share within stated rounding tolerance. PASS.	test_known_answer_base_case
Source / definition	Dated prices cross-checked; Treasury rate checked against official Fed series; cash, investments, debt, leases, and shares reconciled. PASS.	freeze-2026-09-30.md
Three-statement articulation	All forecast-year balance-sheet gaps equal $0.00m; cash flow ties to cash. PASS.	test_three_statement_articulation_and_bridge
DCF monotonicity	Value falls when WACC rises and rises when terminal growth rises. PASS.	test_dcf_monotonicity
Bridge and liquidity	Base bridge and $10bn operating-cash-reserve sensitivity reconcile. PASS.	Bridge test and freeze record
Peer consistency	Pre-existing policy, matching definitions, and GM qualification reviewed. PASS.	lab08.md and tsla_ev_comps.py
Highest-risk failure	Terminal growth at WACC raises ValueError. PASS.	test_highest_risk_terminal_value_failure_fails_loudly
Cold run	python tesla_fcff.py and six-test suite complete without third-party packages for the engine. PASS.	Run recorded October 8, 2026

Locked changed-input record

Field	Student-authored record
Pre-run commitment	October 6, 2026, 7:02 PM; pre-run commit 6b732cd (7:06 PM EDT).
Exact change	FY2027 automotive revenue growth increased from 4% to 6%; all other independent assumptions unchanged.
AI-off prediction	FY2027 FCFF will increase. The watch/defer decision will likely not change because a small driver change cannot close the valuation-market gap.
Actual result	FY2027 FCFF increased from negative $1,770.89m to negative $1,763.49m (+$7.40m). DCF value increased from $21.21 to $21.47 per share (+$0.25).
Reconciliation	Direction matched. Higher FCFF was partly offset by operating working-capital needs; the decision did not change.

Limits and reproducibility

The valuation remains dependent on WACC, terminal growth, cash classification, and a weighted-average share-count convention. Run the engine with python tesla_fcff.py and validation with python -m unittest discover -s tests -p "test_*.py" -v. The semi-adjustable workbench is launched with python -m streamlit run app.py.